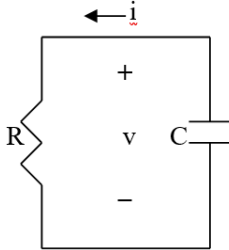


ENGR 2303: Electrical Science

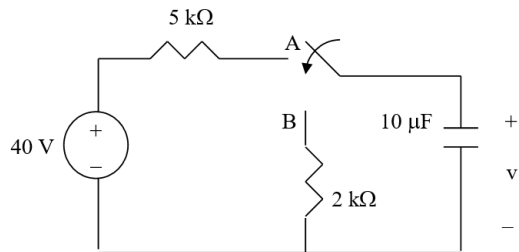
HW 7_Ch7

1) In the circuit shown below $v(t)=56e^{-200t}$ V, $t>0$ $i(t)=8e^{-200t}$ mA, $t>0$

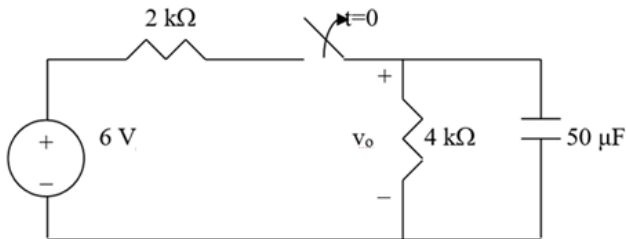
- a) Find the values of R and C. b) Calculate the time constant τ
c) Determine the time required for the voltage to decay half its initial at $t=0$.



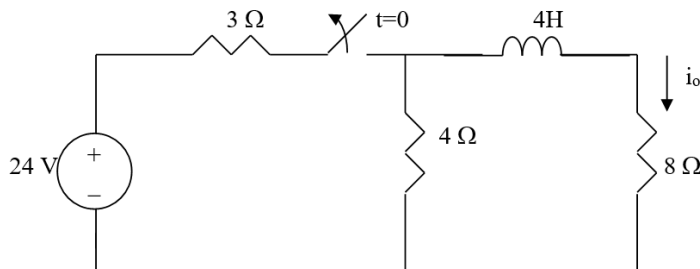
2) The switch in the circuit below moves instantaneously from A to B at $t=0$. Find v for $t>0$.



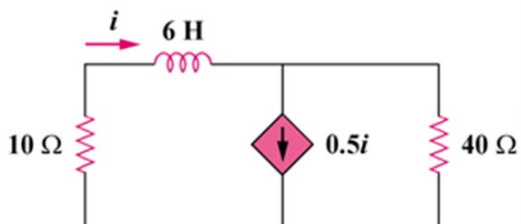
3) The switch in the circuit below opens at $t=0$. Find v_o for $t>0$.



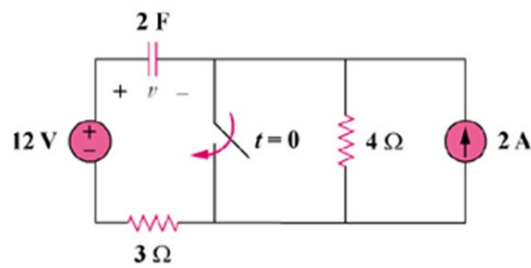
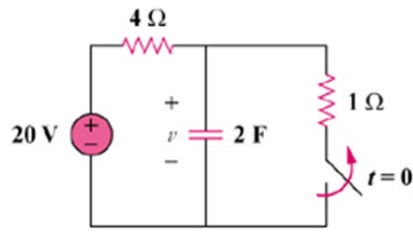
4) For the circuit below, find i_o for $t>0$.



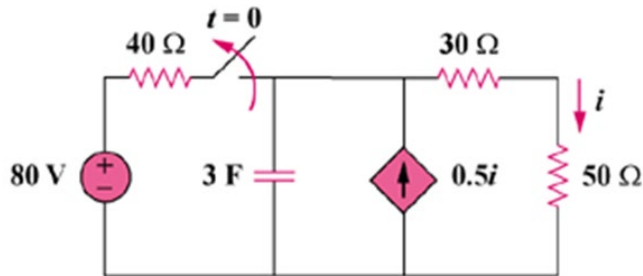
5) In the circuit below, find $i(t)$ for $t>0$ if $i(0) = 2$ A.



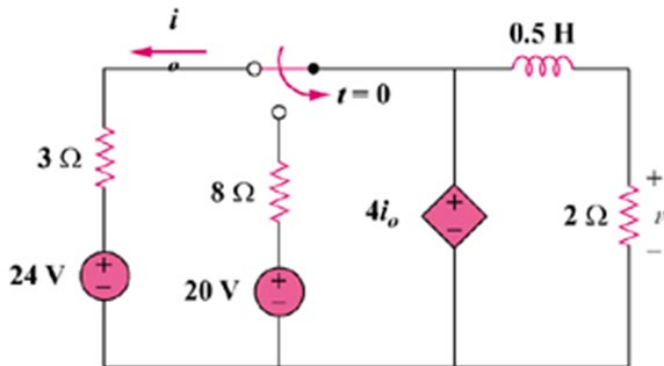
6) Calculate the capacitor voltage for $t < 0$ and $t > 0$ for each of the circuits below.



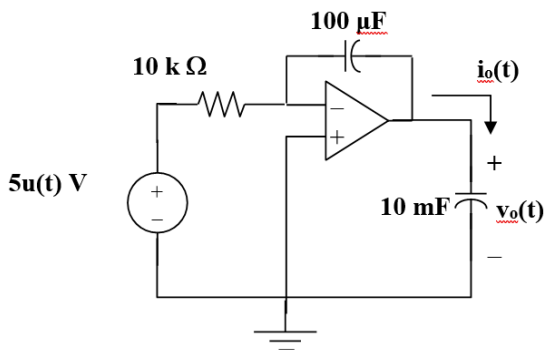
7) Consider the circuit below. Find $i(t)$ for $t < 0$ and $t > 0$.



8) Find $v(t)$ for $t < 0$ and $t > 0$ in the circuit below.



9) For the following circuit, solve for $i_o(t)$.



10) A RC circuit consists of a series connection of a 120-V source, a switch, a 34-MΩ resistor and a 15-μF capacitor. The circuit is used in estimating the speed of a horse running a 4 km racetrack. The switch closes when the horse begins and opens when the horse crosses the finish line. If the capacitor charges to 85.6 V, calculate the speed of the horse.